



NARAYANA
IIT ACADEMY
INDIA

41
YEARS
OF EXCELLENCE

JEE ADVANCED GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

45
YEARS
OF EXCELLENCE

Sec: OSR.IIT_*CO-SC
Time: 3HRS

Date: 07-04-24
Max. Marks: 180

Name of the Student: _____

H.T. NO:

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07-04-24_OSR.STAR CO-SUPER CHAINA_JEE-ADV_GTA-4(P1)_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

TIME:3hrs**IMPORTANT INSTRUCTIONS****Max Marks: 180****MATHEMATICS**

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 4)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 5 – 10)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 11 – 16)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – IV (Q.N : 17 – 19)	Questions with Non-Negative Integer type	+4	0	3	12
Total				19	60

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 20 – 23)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 24 – 29)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 30 – 35)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – IV (Q.N : 36 – 38)	Questions with Non-Negative Integer type	+4	0	3	12
Total				19	60

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 39 – 42)	Questions with Single Correct Options	+3	-1	4	12
Sec – II (Q.N : 43 – 48)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 49 – 54)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – IV (Q.N : 55 – 57)	Questions with Non-Negative Integer type	+4	0	3	12
Total				19	60

MATHEMATICS**MAX.MARKS: 60****SECTION – 1 (Maximum Marks: 12)**

This section contains **FOUR (04)** questions.

Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.

For each question, choose the correct option corresponding to the correct answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If only the correct option is chosen.

Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)

Negative Marks: -1 In all other cases.

- Two rays are drawn a point A at an angle of 30° . A point B is taken on one of them at a distance a from the point A. A perpendicular is drawn from the point B to the other ray and another perpendicular is drawn from its foot to AB to meet AB at another point from where the similar process is repeated indefinitely. The length of the resulting infinite polygon line is:
 A) $a(2-\sqrt{3})$ B) $a(2+\sqrt{3})$ C) a D) $a(\sqrt{3}-1)$
- If $a \sin \theta - b \cos \theta = -\sin 4\theta$ and $a \cos \theta + b \sin \theta = \frac{5}{2} - \frac{3}{2} \cos 4\theta$, then $(a+b)^{2/5} + (a-b)^{2/5}$ is _____.
 A) -5 B) 8 C) 2 D) 6
- Solutions of the equation $\sin^{-1}\left(\left|(\log_6(\cos x))^2 - 1\right|\right) + \cos^{-1}\left(\left|(3 \log_6(\cos x))^2 - 7\right|\right) = \frac{\pi}{2}$, if $x \in [0, 4\pi]$.
 A) $\cos x = 6^{-\sqrt{3}}$ and $6^{-\sqrt{2}}$ B) $\sin x = 6^{-\sqrt{3}}$ and $6^{-\sqrt{2}}$
 C) $\cos x = 6^{-\sqrt{6}}$ and $6^{-\sqrt{5}}$ D) $\sin x = 3^{-\sqrt{6}}$ and $3^{-\sqrt{2}}$
- If $(x-a)\cos\theta + y\sin\theta = (x-a)\cos\phi + y\sin\phi = a$, $\tan\frac{\theta}{2} - \tan\frac{\phi}{2} = 2e$ and θ, ϕ are unequal angles less than 360° , then y^2 is equal to:
 A) $2ax - (1+e^2)x^2$ B) $2ax - (1-e^2)x^2$ C) $2ax + (1-e^2)x^2$ D) $2ax + (1+e^2)x^2$

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.

The answer to each question is a **NUMERICAL VALUE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal places.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +2 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 5 and 6:

Consider $3 \sin \beta = \sin(2\alpha + \beta)$. Let K be the value of $\frac{\tan(\alpha + \beta)}{\tan \alpha}$ and W be the value of the product

$$(\cot \alpha + \cot(\alpha + \beta)) \cdot (\cot \beta - 3 \cot(2\alpha + \beta)).$$

5. The value of K is equal to:
6. The value of $(K + W)$ is equal to:

Question Stem for Question Nos. 7 and 8:

For $0 < x < \frac{\pi}{2}$, let $P_{mn}(x) = m \log_{\cos x}(\sin x) + n \log_{\cos x}(\cot x)$ where $m, n \in \{1, 2, \dots, 9\}$.

[For example : $P_{29}(x) = 2 \log_{\cos x}(\sin x) + 9 \log_{\cos x}(\cot x)$ and

$$P_{77}(x) = 7 \log_{\cos x}(\sin x) + 7 \log_{\cos x}(\cot x)]$$

7. If $P_{34}(x) = P_{22}(x)$, then the value of $\sin x$ is equal to:
8. The mean proportional of numbers $P_{49}\left(\frac{\pi}{4}\right)$ and $P_{94}\left(\frac{\pi}{4}\right)$ is equal to ____

Question Stem for Question Nos. 9 and 10:

Let $f(x) = \sin^{-1} x + |\sin^{-1} x| + \sin^{-1} |x|$.

9. The number of solution(s) of the equation $f(x) = x$ is
10. No. of solution of $f(x) = k$, for true set of values of $K \in \left(0, \frac{\pi}{2}\right]$

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

11. Let $f_n(\theta) = \sum_{n=0}^n \frac{1}{4^n} \sin^4(2^n \theta)$. Then which of the following alternative(s) is/are correct?

A) $f_2\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$ B) $f_3\left(\frac{\pi}{8}\right) = \frac{2+\sqrt{2}}{4}$ C) $f_4\left(\frac{3\pi}{2}\right) = 1$ D) $f_5(\pi) = 0$

12. Let $0 \leq x, y, z \leq \frac{\pi}{2}$ such that $\sin x \sin y \cos z = \frac{1}{2\sqrt{2}}$, $\frac{\sin^2 x \sin^3 y}{\cos z} = \frac{1}{4\sqrt{2}}$ and $\frac{\sin x \sin^4 y}{\cos^2 z} = \frac{1}{8}$, then which of the following is(are) correct?
- A) $\sin x + \sin y = \frac{3}{2}$ B) $\cos(x - y) = \frac{\sqrt{3}}{2}$
 C) $\tan(x + y + z) = \sqrt{3} - 2$ D) $\sec(y - z) = \sqrt{6} - \sqrt{2}$
13. If $(\sin 2x + \sqrt{3} \cos 2x)^2 - 5 = \cos\left(2x - \frac{\pi}{6}\right)$ then the values of $\cos x$ can be:
- A) $\frac{\sqrt{6} + \sqrt{2}}{4}$ B) $\frac{\sqrt{6} - \sqrt{2}}{4}$ C) $-\left(\frac{\sqrt{6} + \sqrt{2}}{4}\right)$ D) $\frac{\sqrt{2} - \sqrt{6}}{4}$
14. Let $S_n(\theta) = \sum_{r=0}^n \frac{2 \sec(2^{r+2}\theta)}{\cot(2^r\theta) - \tan(2^r\theta)}$. Identify the correct statements
- A) $S_3\left(\frac{\pi}{6}\right) = 0$ B) $S_5\left(\frac{\pi}{3}\right) = -\sqrt{3}$ C) $S_5\left(\frac{\pi}{3}\right) = 0$ D) $S_3\left(\frac{\pi}{6}\right) = -2\sqrt{3}$
15. If $\cos^2 \frac{\pi}{9} \cos \frac{4\pi}{9} \cos \frac{7\pi}{9} = \frac{1}{32} \left(\sqrt{b} - \sqrt{a} \cot \frac{\pi}{9} \right)$, $(a, b \in \mathbb{N})$ and $\gcd(a, b) = 1$. Possible values of a, b
- A) a=3 B) a=2 C) b=2 D) b=1
16. Which of the following quantities is/are positive?
- A) $\cos(\tan^{-1}(\tan 4))$ B) $\sin(\cot^{-1}(\cot 4))$ C) $\tan(\cos^{-1}(\cos 5))$ D) $\cot(\sin^{-1}(\sin 4))$

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**. For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

17. If $\sum_{r=1}^{44} \log_2(\sin 2r^\circ) - \sum_{r=1}^{89} \log_2(\cos r^\circ)$, then the value of $\left\lceil \frac{k}{10} \right\rceil$ is, where $\lceil \cdot \rceil$ denotes GIF
18. The sum $\sum_{x=2}^{44} 2 \sin x \cdot \sin 1 [1 + \sec(x-1) \cdot \sec(x+1)]$ can be written in the form as $\sum_{n=1}^4 (-1)^n \frac{\phi^2(\theta_n)}{\psi(\theta_n)}$ where ϕ and ψ are trigonometric functions and $\theta_1, \theta_2, \theta_3, \theta_4$ are in degrees $\in [0, 45^\circ]$. Find the value of $(\theta_1 + \theta_2 + \theta_3 + \theta_4 - 90^\circ)$ in degree
19. If $\theta_1, \theta_2, \theta_3, \dots, \theta_{10} \in \left[0, \frac{\pi}{2}\right]$ such that $\sin^2 \theta_1 + \sin^2 \theta_2 + \dots + \sin^2 \theta_{10} = 1$ then least value of $\left(\frac{\cos \theta_1 + \cos \theta_2 + \dots + \cos \theta_{10}}{\sin \theta_1 + \sin \theta_2 + \dots + \sin \theta_{10}} \right)$ is

PHYSICS**MAX.MARKS: 60****SECTION – 1 (Maximum Marks: 12)**

This section contains **FOUR (04)** questions.

Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer. For each question, choose the correct option corresponding to the correct answer.

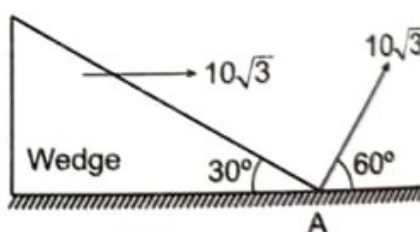
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If only the correct option is chosen.

Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)

Negative Marks: -1 In all other cases.

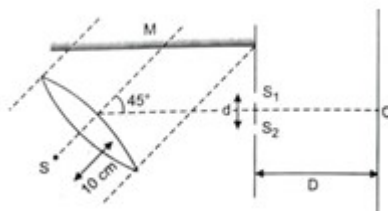
20. A particle is projected at angle 60° with speed $10\sqrt{3} \text{ m/sec}$, from the point 'A' as shown in the figure. At the same time the wedge is made to move with speed $10\sqrt{3} \text{ m/sec}$ towards right as shown in the figure. Then the time after which particle will strike with wedge is : ($g = 10 \text{ m/sec}^2$)



- A) 2 sec B) $2\sqrt{3}$ sec C) $\frac{4}{\sqrt{3}}$ sec D) $\frac{2}{\sqrt{3}}$ sec
21. A spherical ball of mass 4 m , density σ and radius r is attached to a pulley-mass system as shown in figure. The ball is released in a liquid of coefficient of viscosity η and density $\rho \left(< \frac{\sigma}{2} \right)$. If the length of the liquid column is sufficiently long, the terminal velocity attained by the ball is given by (assume all pulleys to be massless and string as massless and inextensible):



- A) $\frac{1}{9} \frac{r^2 (2\sigma - \rho) g}{\eta}$ B) $\frac{1}{9} \frac{r^2 (\sigma - 2\rho) g}{\eta}$ C) $\frac{1}{9} \frac{r^2 (\sigma - 4\rho) g}{\eta}$ D) $\frac{1}{9} \frac{r^2 (\sigma - 3\rho) g}{\eta}$
22. In Young's double slit experiment, distance between the slits is d and that between the slits and screen is D . Angle between principal axis of lens and perpendicular bisector of S_1 and S_2 is 45° . The point source S is placed at the focus of lens and aperture of lens is much larger than d . Assuming only the reflected light from plane mirror M is incident on slits, distance of central maxima from O will be:



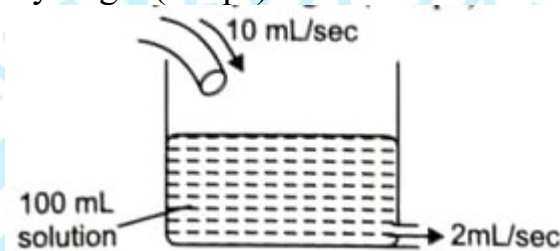
A) $\frac{D}{\sqrt{2}}$

B) $\frac{D}{\sqrt{3}}$

C) $D\sqrt{4}$

D) $\frac{D}{\sqrt{4}}$

23. A 100 mL solution having activity 50 dps is kept in a beaker. It is now constantly diluted by adding water at a constant rate of 10 mL/sec and 2 mL/sec of solution is constantly being taken out. Find the activity of 10 mL solution which is taken out, assuming half life to be effectively very large. (in dps)



A) 50

B) $50\left(\frac{5}{7}\right)^{1/4}$

C) $50\left[1-\left(\frac{5}{7}\right)^{1/4}\right]$

D) $50\left[1+\left(\frac{5}{7}\right)^{1/4}\right]$

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.

The answer to each question is a **NUMERICAL VALUE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal places.

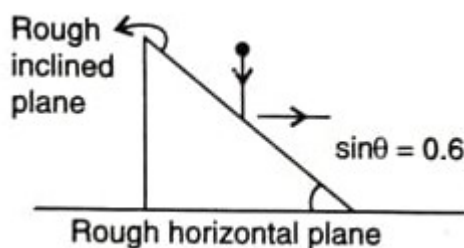
Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +2 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 24 and 25:

A particle falls and strikes the wedge of same mass with a vertical velocity as shown and after collision goes horizontally with half of the striking speed. The coefficient of restitution for this collision is $\frac{9}{16}$, then answer the following questions: (consider $m = 2\text{ kg}$ & $v = 4\text{ m/s}$)

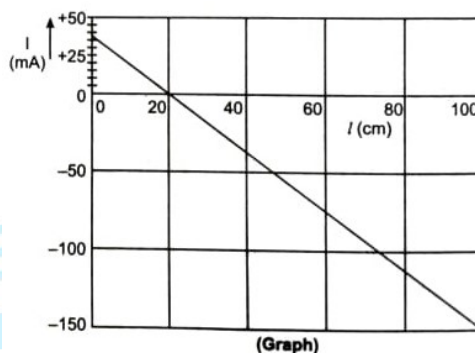
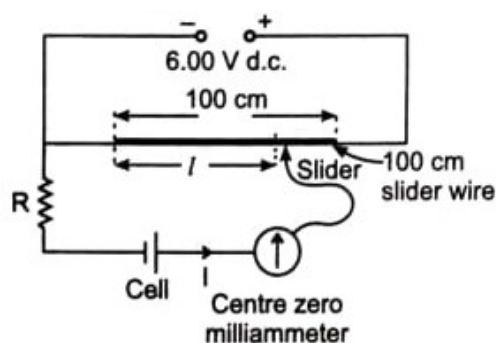


24. The coefficient of friction between the wedge and the horizontal plane is:

25. The K.E of the wedge after collision is: (In joule)

Question Stem for Question Nos. 26 and 27:

In the circuit shown below, the internal resistance of the cell is negligible. The distance of the slider from the left-hand end of the slide wire is l . The graph shows the variation with l of the current I in the cell.



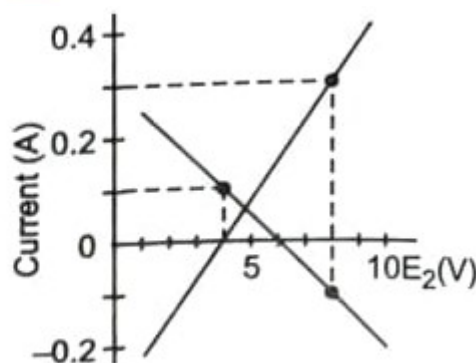
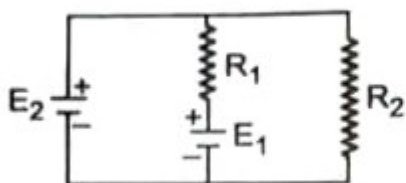
The given ammeter is ideal

26. The balance point is at length l that is equal to: (In cm)

27. Value of the resistance R is:

Question Stem for Question Nos. 28 and 29:

In the circuit shown, both batteries are ideal. EMF E_1 of battery 1 has a fixed value, but EMF E_2 of battery 2 can be varied between 1V and 10V. The graph gives the currents through the two batteries as a function of E_2 , but are not marked as which plot corresponds to which battery. But for both plots, current is assumed to be negative when the direction of the current through the battery is opposite the direction of that battery's emf. (direction from negative to positive)



28. The value of emf E_1 is: (in volt)

29. The resistance R_1 has value : (in ohm)

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

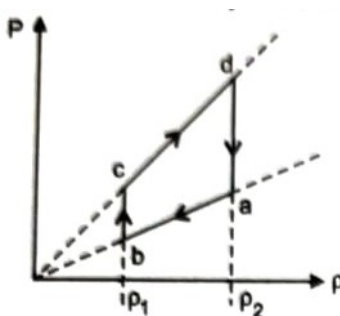
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

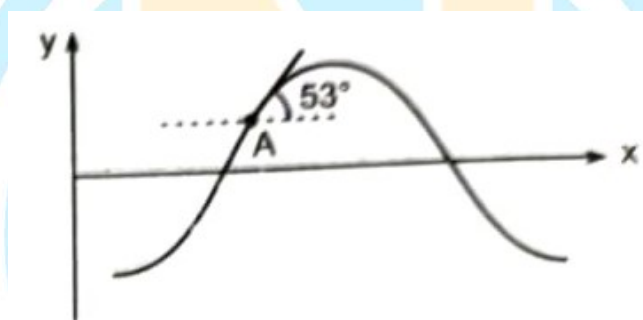
Negative Marks: -2 In all other cases.

30. A particle moves with an initial velocity v_0 and retardation βv , where v is its velocity at any time t (β is a positive constant). Then
- A) the particle will cover a total distance of v_0 / β
 - B) the particle will continue to move for very long time.
 - C) the particle will stop shortly
 - D) the velocity of particle will become $v_0 / 2$ after time $1 / \beta$.
31. Two free point charges $+q$ and $+4q$ are placed a distance x apart. A third charge is so placed that all the three charges are in equilibrium. Then:
- A) unknown charge is $-4q/9$.
 - B) unknown charge is $-9q/4$.
 - C) it should be placed at $(x/3)$ from smaller charge between them .
 - D) it should be placed at $(2x/3)$ from smaller charge between them.
32. Which of the following statements is/are correct about the refraction of light from a plane surface when light ray is incident in denser medium? [C is critical angle]
- A) The maximum angle of deviation during refraction is $\frac{\pi}{2} - C$, it will be at angle of incidence is C .
 - B) The maximum angle of deviation for all angle of incidents is $\pi - 2C$, when angle of incidence is slightly greater than C .
 - C) If angle of incidence is less than C then deviation increases if angle of incidence is also increased.
 - D) If angle of incidence is greater than C then angle if deviation decreases if angle of incidence is increased.

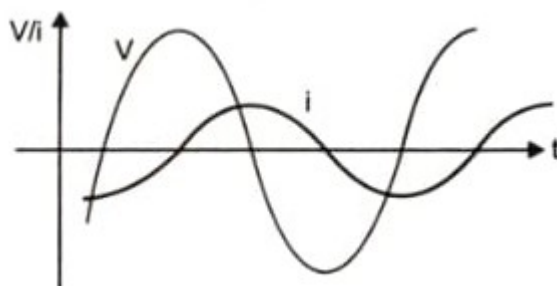
33. An ideal gas undergoes a cyclic process $abcda$ which is shown by pressure-density curve.



- A) Work done by the gas in the process 'bc' is zero.
 B) Work done by the gas in the process 'cd' is negative.
 C) Internal energy of the gas at point 'a' is greater than at state 'c'.
 D) Network done by the gas in the cycle is negative.
34. y - x curve at an instant for a wave travelling along x -axis on a string is shown. Slope at the point A on the curve, as shown, is 53° .



- A) Transverse velocity of the particle at point A is positive if the wave is travelling along positive x -axis.
 B) Transverse velocity of the particle at point A is positive if the wave is travelling along negative x -axis of the particle at point A.
 C) Magnitude of transverse velocity of the particle at point A is greater than wave speed.
 D) Magnitude of transverse velocity of the particle at point A is lesser than wave speed.
35. Graph shows variation of source emf V and current I in a series RLC circuit, with time.



- A) The current leads the emf in the circuit
- B) The circuit is more inductive than capacitive.
- C) To increase the rate at which energy is transferred to the resistive load, L should be decreased.
- D) To increase the rate at which energy is transferred to the resistive load, C should be decreased.

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**

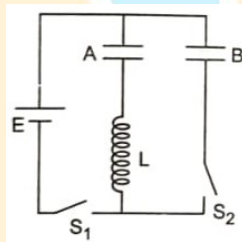
For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

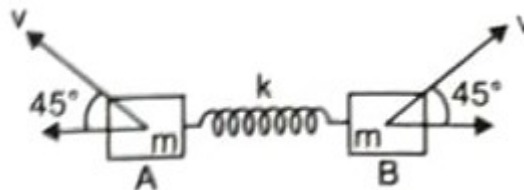
Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

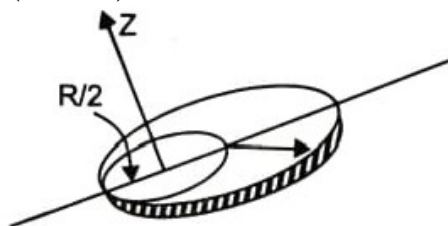
36. Two uncharged identical capacitors A and B, each of capacitance C, and an inductor of inductance L are arranged as shown in the adjacent figure. At $t = 0$, the switch S_1 is closed while switch S_2 remains open. At time $t = t_0 = \sqrt{LC} \frac{\pi}{2}$, switch S_2 is closed while switch S_1 is opened. If the charge on capacitor A at that time is αCE then the value of α is:



37. Blocks A and B of mass m each are connected with spring of constant k . Both block lie on frictionless ground and are imparted horizontal velocity v as shown when spring is unstretched. The maximum stretch of spring is $v\sqrt{\frac{xm}{8k}}$. Then find the value of x .



38. Find the moment of inertia(I) of a thin disc of mass m and radius R with a hole of radius $\frac{R}{2}$ around the Z-axis (see figure), if $m = 6 \text{ kg}$, $R = 2 \text{ m}$ ($m = 6 \text{ kg}$) is the mass of the disc after removing the hole. If $I = (20 + x) \text{ kg m}^2$, where 'x' is



CHEMISTRY**MAX.MARKS: 60****SECTION – 1 (Maximum Marks: 12)**

This section contains **FOUR (04)** questions.

Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer. For each question, choose the correct option corresponding to the correct answer.

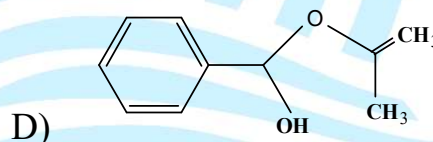
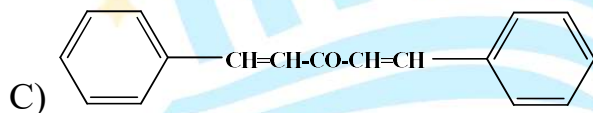
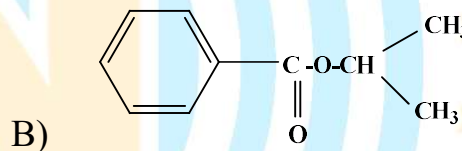
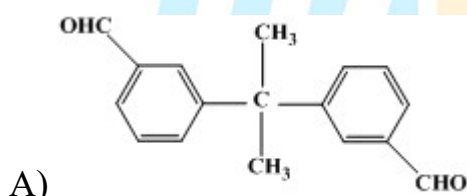
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If only the correct option is chosen.

Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)

Negative Marks: -1 In all other cases.

39. A solid element (monoatomic) exists in cubic crystal. If its atomic radius is $1.0 \text{ \AA}$ and the ratio of packing fraction and density is $0.1 \text{ cm}^3/\text{g}$, then the atomic mass of the element is ($N_A = 6 \times 10^{23}$)
- A) 8π B) 16π C) 80π D) 4π
40. Equal volumes of $M/20$ glucose solution at 300K and $M/20$ sucrose solution at 300 K are mixed without change in temperature. If the osmotic pressure of glucose solution, sucrose solution and the mixture of two solutions are π_1, π_2 and π_3 respectively, then
- A) $\pi_1 = \pi_2 = \pi_3$ B) $\pi_1 > \pi_2 > \pi_3$ C) $\pi_1 < \pi_2 < \pi_3$ D) $\pi_1 = \pi_2 < \pi_3$
41. Which of the following compound is obtained when acetone and benzaldehyde is mixed and heated in the presence of aqueous sodium hydroxide?



42. In the extraction of aluminium
- Process X : employed for red bauxite to remove iron oxide (main impurity)
- Process Y : used for white bauxite to remove Z (main impurity) then,
- Select correct option for the process X and impurity Z.

- A) X = Hall and Heroult's process and Z = SiO_2
- B) X = Bayer's process and Z = SiO_2
- C) X = Serpeck's process and Y = iron oxide
- D) X= Bayer's process and Y = iron oxide

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.

The answer to each question is a **NUMERICAL VALUE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal places.

Answer to each question will be evaluated according to the following marking scheme:

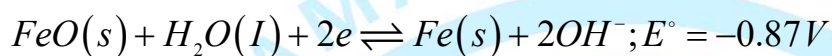
Full Marks: +2 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 43 and 44:

The Edison storage cell is represented as

$Fe(s)|FeO(s)|KOH(aq)|Ni_2O_3(s)|NiO(s)|Ni(s)$. The half-cell reactions are

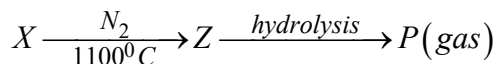
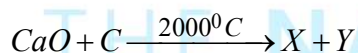


43. What is the standard cell EMF (in volts)?
44. What is the maximum amount of electrical energy (in kJ) that can be obtained from one mole of Ni_2O_3 ?

Question Stem for Question Nos. 45 and 46:

1 mole of a compound P can discharge colour of bromine water and Bayer's reagent. On reaction with $H_2 / Pd - BaSO_4$ (1mole) gives Q (1 mole) which on ozonolysis gives acetaldehyde and glyoxal.

45. Number of carbon atoms in P is _____
46. Degree of unsaturation of Q is _____

Question Stem for Question Nos. 47 and 48:

47. Number of carbon atoms in X is
48. How many of the following can dissolve gas P?

Water, Aq. HCl solution, Aq NaOH, *dil.* H_2SO_4 , benzene, CCl_4

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

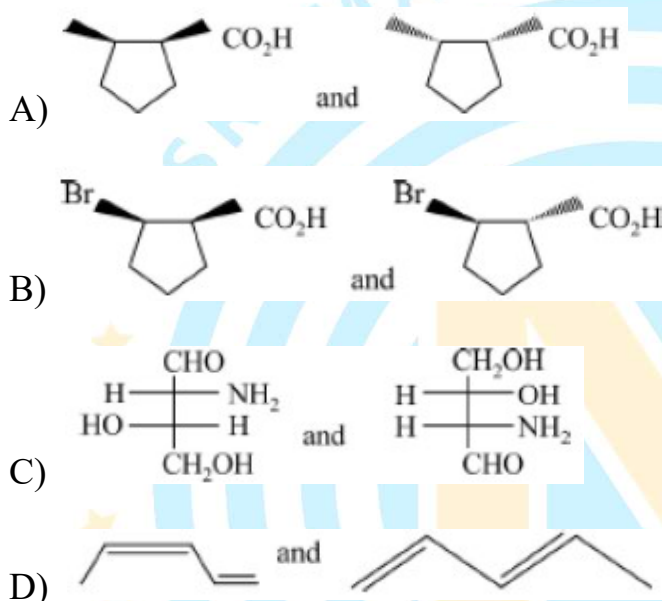
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

49. The diastereomeric pairs are



50. The rate law for the reaction



The rate of the reaction will be

A) doubled on doubling the concentration of NaOH.

B) halved on reducing the concentration of alkyl halide to one half.

C) increased on increasing the temperature of the reaction.

D) unaffected by increasing the temperature of the reaction.

51. The reaction(s) which does (do) not occur in the reduction zone in the extraction of iron from haematite ore is (are) :
- A) $Fe_2O_3 + CO \rightarrow 2FeO + CO_2$ B) $FeO + CO \rightarrow FeO + CO_2$
C) $Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$ D) $CaO + SiO_2 \rightarrow CaSiO_3$
52. A complex of certain metal has the magnetic moment of 4.91 BM whereas another complex of the same metal with same oxidation state has zero magnetic moment, except
- A) Co^{+2} B) Mn^{+2} C) Fe^{+2} D) Fe^{+3}
53. A cylinder fitted with a movable piston contains liquid water in equilibrium with water vapour at 25°C. Which operation results in a decrease in the equilibrium vapour pressure?
- A) Moving the piston downwards a short distance.
B) Removing a small amount of vapour.
C) Removing a small amount of the liquid water.
D) Dissolving salt in the water.
54. Addition of excess aqueous ammonia to a pink coloured aqueous solution of $MCl_3 \cdot 6H_2O(X)$ and gives an octahedral complex Y in the presence of air. In aqueous solution, complex Y behaves as 1:3 electrolyte. The reaction of X with excess HCl at room temperature results in the formation of a blue coloured complex Z. The calculated spin only magnetic moment of X and Z is 3.87 BM, whereas it is zero for complex Y. Among the following options, which statements is(are) CORRECT ?
- A) The hybridization of the central metal ion in Y is $d^2 sp^3$
B) Z is tetrahedral complex
C) Addition of silver nitrate to Y gives only two equivalents of silver chloride
D) When X and Y are in equilibrium at 0°C, the colour of the solution is pink

SECTION - 4 (Maximum Marks : 12)

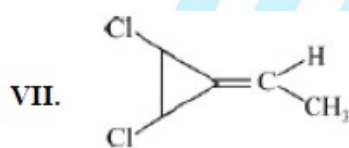
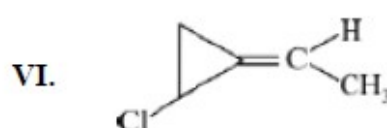
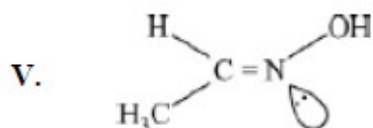
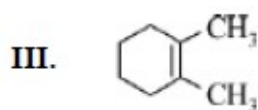
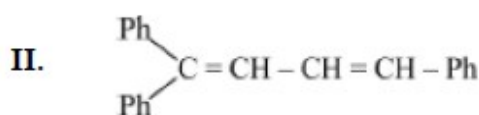
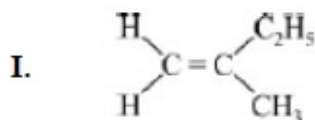
This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**. For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

55. How many of the following shows geometrical isomerism?



56. The Van der Waals constants *a* and *b* for a gas of molar mass 164.2 g/mol are 4.105 atm-L²/mol² and 0.1 L/mol, respectively. The density (in kg/m³) of the gas at 2 atm and 500 K is

57. During roasting of galena, a gaseous substance is evolved, how many of the following are correct for this gaseous substance?

i) It causes temporary bleaching of moist red litmus paper

ii) It turns acidified $K_2Cr_2O_7$ solution to green

iii) It is one of the major contributors of acid rain

iv) It can act as both oxidizing and reducing agent.